



GUWE HIGH SCHOOL EXAMINATIONS

General Certificate of Education Ordinary Level

MATHEMATICS

4004/1

PAPER 1

JULY 2024 MID YEAR EXAMINATIONS (F3)

Additional materials;

- Answer scripts

Allow candidates 5 minutes to count pages before the examination.

TIME **2 hours 30 minutes**

INSTRUCTIONS TO CANDIDATES

Write Name and surname on each and every answer sheet which you have used.

Check that the question paper has all the pages and ask the invigilator for help if there are duplicates or if there are missing pages.

Answer all questions

Write your answers using blue or black pens

Show all your workings

Omission of essential working will result in loss of marks.

Decimals answers which are not exact should be given correct to three significant figures unless stated otherwise.

Mathematical tables, slide rules and electronic calculator should not be brought into the examination room.

INFORMATION FOR CANDIDATES

The number of marks is given in [] at the end of each and every question or part question.

The total number of marks for this paper is 100.

You are reminded of need of clear presentations in your answers.

This question paper consists of 5 printed pages and 1 blank page

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EXODUS 14: 13-18~MUGS

1. Express 68,978 to
 - a) the nearest 10. [1]
 - b) the nearest tenth. [1]
 - c) one significant figures. [1]

2. Find the exact value of
 - a) $0,04 \times 0,3$ [1]
 - b) $1,44 \div 0,09$ [1]
 - c) $0,89 - 1,72$ [1]

3. (a) Find the value of $\frac{8}{0,04}$ [1]
 (b) Simplify $1\frac{1}{2} - \frac{4}{7} \div \frac{2}{3}$ [2]

4. Given that $p = -4$, $q = 3$ and $r = 1$. Evaluate
 - a) $\frac{p+q}{r}$ [1]
 - b) $\sqrt{p^2q - r}$ [2]

5. In an athletics competition, under 20 boys compete in 5 000 m race, while under 17 boys compete 3 000 m race.
 - a) A lap is 400m long. Find the number of laps in the 5 000m race. [1]
 - b) Calculate the difference between the distance for under 20 competitions and that of under 17 competitions giving your answer in standard form. [2]

6. (a) Express $1 \times 3^3 + 2 \times 3^2 + 1 \times 3$ in base 3. [1]
 (b) Convert 101_{10} to number in base 9. [1]
 (c) Evaluate $203_7 - 154_7$ in base 7. [1]

7. Solve the simultaneous equations

$$\begin{aligned} 2x + 3y &= 28 \\ x + 5y &= 35 \end{aligned}$$

[3]

8. Solve $\frac{2y+5}{3y-2} = \frac{9}{4}$ [3]

9. If $\log 7 = 0,845$. Evaluate

a) $\log 49$ [1]

b) $\log 0,07$ [1]

c) $\log \sqrt{7}$ [1]

10. Evaluate

a) $81^{\frac{3}{4}}$ [1]

b) $(\frac{32}{50})^{-\frac{1}{2}}$ [2]

11. (a) Evaluate $\log_6 36$ [1]

(b) Find x , if $9^{x-1} \times 3^{3x-2}$ [2]

12. Factorise completely

a) $x^2 - \frac{1}{36}$ [1]

b) $5y^2 - y - 4$ [2]

13. Express as a fraction in its lowest terms

$$\frac{4}{x+5} - \frac{3}{x} \quad [3]$$

14. (a) Show that $\frac{6x}{5} + \frac{5}{x} = 3x$ reduces to $9x^2 - 25 = 0$ [2]

(b) Hence or otherwise solve $\frac{6x}{5} + \frac{5}{x} = 3x$ [2]

15. Factorise

a) $10m^2 - tr - 2m + 5mr$ [2]

b) $p^3 - 16p$ [2]

16. Solve the following equations

a) $(3x - 2)(x + 1) = 0$ [2]

b) $(y - \frac{2}{9})^2 = \frac{49}{81}$ [2]

17. (a) Remove brackets and simplify $(2x - y)(3x + 2y)$ [2]

(b) Express 0,8125 as a fraction in its lowest terms. [2]

18. Given that $p = \frac{q}{q+m}$.

- a) Make m the subject of formula. [2]
b) Find m if $q = 3$ and $p = 2$. [2]

19. Evaluate

- a) $\frac{\log 8 - \log 4}{\log 4 - \log 2}$ [3]
b) $\log_2 \frac{1}{32}$ [2]

20. Given that $f(x) = x^2 - 3x - 10$. Find

- a) $f(0)$ [1]
b) $f(-1)$ [1]
c) x if $f(x) = 0$ [3]

21. It is given that $s = ut - \frac{1}{2}gt^2$.

- i) Find the value of s if $g = 9,8$; $u = 20$ and $t = 2$ [2]
ii) Make g the subject of formula. [3]

22. Find the

- a) Lowest common multiple of 18 and 24. [2]
b) Highest Common Factor of 168 and 252. [3]

23. Simplify

- a) $\frac{1}{a^2-3a+2} \div \frac{1}{1-a}$ [3]
b) $\frac{m^2-mn}{n^2-np} \div \frac{m}{n-p}$ [2]

24. (a) Factorise completely

- (i) $x^2 + 2x - 3$ [2]
(ii) $x^2 - 1$ [1]

(b) Hence or otherwise find the HCF of $x^2 + 2x - 3$ and $x^2 - 1$. [2]

25. Find the value of x if

a) $111_x = 7_{10}$ [3]

b) $x_8 + 234_5 = 421_5$ [4]

26. The cost of 3kg of apples and 7kg of bananas is \$16.

The cost of 4kg of apples and 5kg of bananas is \$17.

Calculate the cost per kg of

a) apples [2]

b) bananas [2]

ALL THE BEST!!

Exodus 14: 13..... Fear not, stand firm, and see the salvation of the LORD, which he will work for you today. For the Egyptians whom you see today, you shall never see again.

u-Eksodusi 14:13:Lingesabi; manini liqine, libone usindiso lwe Nkosi ezalenzela lona lamuhla; ngoba amaGibhithe eliwabonayo lamuhla kalisayikubona futhi ephakadeni.